

Nitish Shukla

PhD Student, Computer Science (Computer Vision & Multimodal Learning)

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Portfolio — Google Scholar — GitHub — LinkedIn

Research Overview

My research lies at the intersection of representation learning and model reliability in deep learning, with a focus on how complex neural systems encode, reason over, and sometimes fail to disentangle fine-grained visual information in security-critical settings. I study how multimodal large language models (MLLMs) encode and reason over facial information, analyze failure modes in multi-image reasoning, and leverage internal representations for robust visual inference. A central theme of my work is generative face demorphing, where I develop models, benchmarks, and evaluation protocols that improve both reconstruction quality and downstream morph attack detection.

Education

PhD in Computer Science Michigan State University, MI, USA	Jan 2024 – Present
MSc. in Computer Science Chennai Mathematical Institute, Chennai, India	Dec 2020 – Jun 2022
MSc. in Mathematics Indian Institute of Technology Guwahati, India	Jul 2017 – Jun 2019
BSc. in Physics, Mathematics and Computer Applications Kanpur University, India	Jul 2014 – Jun 2017

Employment

Adobe Research — PhD Research Intern Developed an attribute-aware style validation framework using MLLM-induced embeddings for fine-grained artistic style analysis and retrieval. Proposed an attribute-based image decomposition strategy that improved Recall@K by 2–3% over baseline representations. (<i>Manuscript under review at ACL 2026</i>).	May 2025 – Aug 2025
Micron Technology — Data Scientist II Designed a synthetic data generation pipeline to fine-tune a proprietary defect segmentation system for semiconductor die inspection. Synthetic augmentation improved defect shape mIoU by 9% , enabling robust detection of rare failure modes.	Aug 2022 – Aug 2023

Ongoing Research Projects

Enhancing Single-Image Facial Demorphing using MLLMs

Leveraged MLLM-derived semantic representations of morphs to condition a custom image-to-image diffusion model, achieving a 2–3% improvement in face matching accuracy over baseline demorphing methods. (*Submitted, T-PAMI, Feb 2026*).

Desired Difficulty for Data-Efficient Multi-Image Preference Optimization

Proposed a data-efficient multi-image preference optimization framework inspired by desirable difficulties. Designed a prompt-driven data generation pipeline exploiting intrinsic inter-image relationships, reducing annotation cost by **60%** while improving performance by **5–6%** across multi-image benchmarks. (*Under review, ACL 2026*).

Style-Aware Open-Set Image Retrieval via MLLMs

Introduced an open-set image retrieval framework leveraging MLLM zero-shot reasoning to extract attribute-disentangled representations (e.g., brushstrokes, color palette, composition). Achieved **2–3% mAP** gains without supervised training. (*Under review, ACL 2026*).

Publications

1. **Nitish Shukla**, Arun Ross. *Facial Demorphing from a Single Morph Using a Latent Conditional GAN*. In Proceedings of IEEE International Joint Conference on Biometrics (IJCB), 2025. (**IAPR Best Student Paper Award**)
2. **Nitish Shukla**, Arun Ross. *diffDemorph: Extending Reference-Free Demorphing to Unseen Faces*. In Proceedings of IEEE International Conference on Image Processing (ICIP), 2025.
3. **Nitish Shukla**, Arun Ross. *dc-GAN: Dual-Conditioned GAN for Face Demorphing From a Single Morph*. In Proceedings of IEEE International Conference on Automatic Face and Gesture Recognition (FG), 2025.
4. **Nitish Shukla**, Arun Ross. *Metric for Evaluating Performance of Reference-Free Demorphing Methods*. In Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW), 2025.
5. **Nitish Shukla**, Arun Ross. *Face Demorphing via Identity Preserving Image Decomposition*. In Proceedings of IEEE International Joint Conference on Biometrics (IJCB), 2024.
6. **Nitish Shukla**, Sudipta Banerjee. *Generating Adversarial Attacks in Latent Space*. In Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2023.
7. **Nitish Shukla**. *SDeMorph: Towards Better Component Face Extraction from Single Morph*. In Proceedings of IEEE International Joint Conference on Biometrics (IJCB), 2023.

Preprints

1. **Nitish Shukla**, Arun Ross. *Enhancing Single-Image Facial Demorphing via Multimodal Large Language Models*. preprint, 2025.
2. **Nitish Shukla**, Sargan Jandial, Arun Ross. *S2H-DPO: Hardness-Aware Preference Optimization for Vision–Language Models*. Under Review, 2025.
3. **Nitish Shukla**, Siddhant Prakash, Arun Ross, Harsh Rangwani. *Harnessing Multimodal LLMs for Attribute Specific Image Retrieval*. Under review, 2025.

Technical Skills

Computer Vision: Face biometrics, demorphing, morph attack detection, image retrieval

Deep Learning: GANs, diffusion models, transformers, contrastive learning

Multimodal Models: CLIP, MLLMs, vision-language representation learning

Frameworks: PyTorch, OpenCV, Scikit-learn

Experimentation: MLflow, Pandas, large-scale benchmarking

Mathematics: Bayesian methods, graph theory, linear algebra

References

Arun Ross (Michigan State University) — rossarun@msu.edu

K. V. Subramanayam (Chennai Mathematical Institute) — kv@cmi.ac.in

Sudipta Banerjee (University of Wyoming) — sbanerj3@uwyo.edu